

# Ayush Sunil Munot

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## EDUCATION

Indian Institute of Technology Kharagpur, India

2021 – 2026

B.Tech. in Mechanical Engg., M.Tech. in Financial Engg. (5Y); Minor: CSE; Micro Spl.: AI

CGPA: 8.97 / 10

## PUBLICATIONS

MVEB: Massive Video Embedding Benchmark | Accepted, *EMNLP 2026 Findings* | [arXiv](#)  
SurakshaEval: An Indic Safety Benchmark for Multilingual LLMs | Accepted, *AIES 2026* | [arXiv](#)  
MAEB: Massive Audio Embedding Benchmark | Under Review, *NeurIPS 2026* | [arXiv](#)  
Multilingual GSM-Symbolic: Determining capability transfer across languages? | Under Review, *ICLR 2027*  
TDDN: Text-aligned Diffused DINO Network for Puzzle Understanding | Under Review, *AAAI 2026* | [arXiv](#)  
BenchGuard: Verifying Reward Integrity in LLM Agent Evaluation | Under Review, *USENIX Security '27*  
Workshop papers: Last Translation Benchmark ([arXiv](#)) & MultiCulturalRiddle (Accepted, MRL-EMNLP 26)

## EXPERIENCE

Member of Technical Staff | *Domyn* | Remote

July 2026 – Present

- Building **guardrails** & **evaluation** frameworks for **AI safety** in financial & sovereign agentic deployments

AI Intern | *Fidelity Investments* | Bangalore, India

May 2025 – July 2025

- Built a retrieval-based agent assist system for email/chat/call workflows, reducing query **resolution time** by **30%**
- Indexed **2300+** documents using **FAISS**, built **SQLite** feedback loops, and deployed via Streamlit + **FastAPI**
- Achieved **90%+** top-K retrieval accuracy using **HyQE**, outperforming RAG (**60%**) on a 10k-query test set

ML Engineering Intern | *Sourcegraph* | Remote

May 2024 – Jan 2025

- Improved multi-language unit test generation using **Dspy**, achieving **~10%** gains on **HumanEvalPack** dataset
- Built a **multi-agent** system to process **100k+** Cody logs, enabling dataset creation, fine-tuning, & evaluation
- Synthesized retrieval datasets, integrated **10+** code-retrieval benchmarks, evaluated Cody against **5+** competitors

Research Assistant | *AFAR Lab, University of Cambridge* | Remote

July 2023 – Jan 2024

- Constructed temporal dataset from **OpenFace** features, mapping to frequency-aligned **spectral** representations
- Transformed temporal data into spectral graphs for **GNN**-based multi-class classification with **early & late** fusion
- Achieved **85%** accuracy (**GAT**) & **55%** (**GCN**), surpassing **45%** ML baseline using **Optuna** for parameter tuning

Computer Vision Intern | *Spectrum Lab, IISc Bangalore* | Remote

May 2023 – Aug 2023

- Developed **STAPLECAM**, an **EM**-based endoscopic segmentation method integrating multiple CAM techniques
- Finetuned **ResNet-152** on endoscopic data with cross-entropy loss & **Otsu**-based binarization using 20% test split
- Achieved Dice/IoU scores of **0.86/0.92** on the KVASIR dataset & **0.74/0.85** on the BKAI-IGH-NeoPolyp dataset

## PROJECTS

Multimodal AI Notes Platform (PWA) | *self-project* | [GitHub](#)

Feb 2026 – Mar 2026

- Built a multimodal **PWA** for note-taking using **FastAPI** & **Supabase**, supporting text, audio, & image ingestion
- Designed a pipeline with **Gemini**-based parsing, **pgvector** semantic search, & secure storage via signed URLs
- Shipped **Gemini 2.5 TTS** summaries, deep-link sharing, & **MCP** integrations for AI-assisted knowledge access

Multi-Agent AI for Interactive Physics Simulations | *self-project* | [GitHub](#)

July 2025 – Aug 2025

- Built a self-correcting **multi-agent** system for physics simulation generation, reducing prototyping time by **80%**
- Integrated code **sandbox** with automated dependency management, achieving **90%** autonomous execution rate
- Expanded into a **multimodal** learning platform with **context-aware** chat, voice/image inputs for faster iteration

Adobe Behaviour Simulation Challenge - Gold | *Inter IIT TECH Meet – 12.0*

Nov 2023 – Dec 2023

- Designed **MLP+Transformer** head on **BERT** with **SAM** losses & **PCA**-based inference for tweet-like prediction
- Built supervised routing to optimally select among **3** fine-tuned LLMs; used **CLIP** for enhanced tweet generation
- Achieved **2262.33 RMSE** predicting likes; surpassed **GPT-4** in content prediction (**0.24 BLEU**, **0.25 ROUGE**)

## TECHNICAL SKILLS

Languages: C, C++, Python, SQL, Bash

ML & DL: NumPy, Pandas, SciKit-learn, TensorFlow, PyTorch, OpenCV, Optuna, Matplotlib, Transformers

Data & MLOps: FAISS, PostgreSQL, Redis, AWS, GCP, Docker, MLflow, FastAPI, Git, CI/CD, Streamlit, Gradio

NLP & GenAI: GNNs, Hugging Face, LLM Fine-tuning & Inference, RAG, Agentic AI, LangChain, LangGraph, MCP

## AWARDS & ACHIEVEMENTS

- Core Contributor** in **MTEB** (Massive Text Embedding Benchmark), a leading evaluation repository
- Served as Reviewer**: ACL 2026 (Industry Track), COLM 2026 (CBW Workshop), CVPR 2026 (MMRAGI Workshop) & ICML 2026 (DL4C Workshop); **PC Member** for AAAI/ACM AIES 2026
- Guardian** on LeetCode (top **1.13%**), Max Rating **2211** (Ayush-1234); Global Rank **102** in Weekly Contest 432

## COURSEWORK

Maths & CS: Linear Algebra, Probability & Statistics, Time Series, Data Structures & Algorithms, Complex Networks

Machine Learning: ML Foundations, NLP, Computer Vision; **MOOCs**: CS229, CS230, CS231n, CS224n, CS336